

Question_6:

Code:

```
1  /*
2  QUESTION_6:
3  Create a linked list and find the data stored at a given index.
4  The function required is:
5  int GetNth(struct node* head, int index)
6  The linked list should be displayed first,
7  followed by the node present at the given index.
8  */
9  #include <iostream>
10 using namespace std;
11
12 struct node {
13     int data;
14     node *next;
15 };
16 int GetNth(node *head, int index) {
17     for (int i = 0; i < index; i++)
18         head = head->next;
19     return head->data;
20 }
21
22 int main() {
23     int n, x, index;
24     cin >> n;
25
26     node *head = NULL;
27     for (int i = 0; i < n; i++) {
28         cin >> x;
29
30         node *p = new node{x, head};
31         head = p;
32     }
33     cin >> index;
34     cout << "Linked list:-->";
35     node *p = head;
36
37     while (p) {
38         cout << p->data;
```

```
39         if (p->next)
40             cout << "-->";
41         p = p->next;
42     }
43     cout << endl;
44
45     cout << "Node at index=" << index << ":"
46         << GetNth(head, index);
47     return 0;
48 }
```

Output:

Test Case_1:

```
4
1 2 3 4
2
Linked list:-->4-->3-->2-->1
Node at index=2:2
PS C:\Users\Prabin Yadav\Documents
```

Test Case_2:

```
4
11 21 31 41
3
Linked list:-->41-->31-->21-->11
Node at index=3:11
PS C:\Users\Prabin Yadav\Documents
```